LUCY RANDEWICH

Full-stack software engineer with experience building production insurance systems, cyber-security research tooling, and machine learning applications. Comfortable owning projects from design through deployment across backend, frontend and cloud infrastructure.

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Experience

Software Engineer — Laka *2024 – Present*

- Develop and maintain production systems across Laka's insurance platform, including claims tooling, customer onboarding systems, partner APIs and monitoring infrastructure.
- Led a company-wide claims data remediation project, introducing monitoring and alerting that reduced claim customer support issues to near zero.
- Using AI-assisted development tools to accelerate implementation while maintaining appropriate review, testing and security standards.

Research Associate — Bristol Cyber Security Group *2023 – 2024*

- Produced professional tooling for REPHRAIN's ambitious "Testbed OS" framework while aiding various other research projects in the cyber-security field.
- Designed and implemented a cyber-security training exercise adopted for training sessions delivered by the Metropolitan Police and City of London Police.

Teaching Assistant — University of Bristol *2022 – 2023*

- COMS20006: Software Project Engineering. Led weekly stand-up meetings for students and provided full-stack technical support. Was lead TA for the resit period, helping students who needed more explicit guidance with any aspect of software development.
- COMS20012: Computer Systems B. Taught concepts involving software security and operating systems, which involved instructing at lab sessions and preparing technical worksheets.

Machine Learning Intern — Oxford Nanopore Technologies *2022*

- Validated design choices and optimised neural networks in bioinformatic applications, focusing on a tool for classification of methylated bases.
- Researched differentiable Neural Architecture Search methods for automated architecture tuning.
- Developed pipeline automation tools to allow the MLOps team to train and evaluate models more efficiently.
- Created genome data sets for regions of clinical interest to allow faster evaluation of base calling.

Education

BSc Computer Science — University of Bristol *2020-2023*

- First class honours achieving 79% in Year 2, and a final overall grade of 75%.
- Awarded the 'Hargrave Scholarship' for placing top of the cohort in first and second years of study.
- Received the' 2022 Netcraft prize' for placing in the top 10 second-year students.
- Thesis: Deep Metric Learning for the Visual Identification of Individual Cattle from Depth Imagery (published on arXiv)

The Abbey School, Reading *2013-2020*

- A-Levels: Maths/Further Maths (A*), Computer Science (A*), Physics (A)
- 10 GCSEs at A*

Research & Publications

Deep Metric Learning for the Visual Identification of Individual Cattle from Depth Imagery

- Published research proposing a novel deep metric learning approach for identifying individual cattle from depth imagery. Supervised by Dr. Tilo Burghardt, the published paper can be found at: `https://arxiv.org/abs/2404.00172`
- Curated a new dataset and purpose designed a network architecture for the task of identifying individual cattle from depth imagery alone.
- Evaluated data augmentation, loss functions, and model backbones and performed ablation studies to facilitate interpretation and discussion of quantitative results.
- Performed spatial localisation of inter-class individuality using a Grad-CAM inspired algorithm.

Core Technologies

Languages	TypeScript, JavaScript, Python, Go, SQL
Frameworks	React, Node.js
Cloud	AWS, Docker, CI/CD, GitHub Actions
DevOps	Grafana, Sentry, Argo
Machine Learning	PyTorch, TensorFlow, OpenCV